QuestionCraft — Phase 2 Report

Productionization & Deployment · RV College of Engineering · AI254TA Machine Learning Operations (5th Sem) · November 2026

1. Scope

Phase 2 covers eight deliverables. Kubernetes and statistical drift monitoring are out of scope and have been removed from the repository; Docker Compose is the deployment target.

Area	In Phase 2?	Where
Model testing and data validation	In	src/monitoring/model_tests.py, tests/
Model-risk and quality gates	In	src/monitoring/quality_gates.py
Model registry and promotion	In	src/registry/model_manager.py, scripts/promote_model.py
FastAPI inference service	In	src/serving/
Deployment strategies (blue-green, canary)	In	Registry + API (no Kubernetes)
11 ML artifacts	In	src/registry/artifacts.py, GET /artifacts
Observability (Prometheus, Grafana)	In	/metrics, monitoring/
Docker and Compose	In	Dockerfile, docker-compose.yml
CI/CD	In	.github/workflows/ci.yml
Kubernetes	**Out**	Removed
Drift monitoring	**Out**	Removed

2. Model testing and data validation

Rule-based checks implemented in the project (Great Expectations is not used).

- Column existence, null ratio ≤ 1%, allowed label set, minimum question length 5, row count 100–500,000, feature mean bounds, no duplicate rows.
- **Cleaning before splitting** (clean_text_dataset): the processed difficulty data fails the raw checks (22 short texts, 87,772 duplicate rows); cleaning takes it from 191,775 to 103,989 rows and all 8 checks pass.
- src/data/build_splits.py trains on public/proxy data and holds out real RVCE exam papers; any text that appears in the held-out set is removed from training.

Done when: pytest tests/ passes and the pipeline's data-validation step reports all checks passed.

3. Model-risk and quality gates

No model reaches Production unless every gate passes. Metrics come from the candidate's real MLflow run and are compared with the current Production version.

Gate	Rule
Minimum accuracy	≥ 0.85
Minimum F1-macro	≥ 0.30
No regression	F1 drop vs Production ≤ 2.0%
Stability	5-fold CV std ≤ 0.05
Reproducibility	Training-data MD5 matches Production's
Sample size	≥ 100 test samples

Done when: a model that fails any gate is refused promotion — covered by TestPromotionIsRefused, which runs the promotion CLI against a temporary registry and checks Production is unchanged.

4. Model registry and promotion

- MLflow Model Registry (sqlite:///mlflow.db): None → Staging → Production → Archived.
- Each version is tagged with data_hash, data_file, model_type, run_name and key metrics.
- CLI: --list, --to staging|production (gated), --compare A B, --rollback, --sync-tags.

Done when: a model can be registered, staged, promoted, compared and rolled back from the CLI.

5. FastAPI inference service

Endpoint	Purpose
GET /health	Status, uptime, loaded models
POST /predict, /classify, /batch/classify	Difficulty, Bloom's level, CT skills
POST /dedup, /batch/dedup	Semantic duplicate check
GET /models/info	Serving versions, canary version and ratio, registry stages
POST /models/reload	Pick up a promotion or rollback without restarting
GET /artifacts	The 11-artifact manifest
GET /deploy/status, POST /deploy/canary	Deployment strategy and canary ratio
GET /metrics	Prometheus text format

Done when: all endpoints answer, input is validated, and tests/test_serving.py passes.

6. Deployment strategies

- **Blue-green:** Production serves 100%; the Staging candidate is promoted only after the gates pass; POST /models/reload swaps it in; --rollback restores the previous version.
- **Canary:** ModelStore routes a configurable share of requests (0.0–1.0) to the Staging model; POST /deploy/canary?ratio=0.1 sets 10%.
- The evaluator demo drives the real API and prints /models/info at baseline, canary 10%, after promotion (v6 → Production) and after rollback (v5 restored).

Done when: a demo shows a canary at 10%, a promotion and a rollback, with /models/info reflecting each step.

7. The 11 ML artifacts

Defined once in src/registry/artifacts.py, served by GET /artifacts and verified by the pipeline.

#	Artifact	Location
1	Dataset	DVC-versioned Parquet in data/processed/ (+ data/splits/)
2	Preprocessing pipeline	src/data/preprocess.py, src/data/build_splits.py
3	Trained models	models/*.pkl (difficulty, Bloom's, CT)
4	Model configuration	configs/model_config.yaml
5	Evaluation results	MLflow metrics, incl. rvce_* held-out metrics
6	Training logs	MLflow run telemetry
7	Feature engineering	src/data/feature_engineering.py
8	Tokenizer / vectorizer	TF-IDF saved inside each model pipeline
9	Model metadata	MLflow registry version tags
10	Deployment artifacts	Dockerfile, docker-compose.yml, CI workflow
11	Monitoring artifacts	monitoring/prometheus.yml, Grafana dashboard + provisioning

8. Observability

Prometheus scrapes /metrics every 10 s; Grafana shows a provisioned dashboard on port 3000. Observability covers service health and traffic only.

- questioncraft_requests_total, questioncraft_request_latency_seconds_total, questioncraft_predictions_total, questioncraft_models_loaded, questioncraft_uptime_seconds.

9. Docker

- Multi-stage Dockerfile (python:3.13-slim); runtime dependencies pinned to the versions the models were trained with (requirements-serving.txt); CPU-only torch; one Uvicorn worker; HEALTHCHECK.
- docker-compose.yml: API :8000, MLflow :5000, Prometheus :9090, Grafana :3000. API and MLflow share the host registry so the API serves Production and can canary Staging.

Done when: docker compose up -d brings up all four services healthy and /docs loads on port 8000.

10. CI/CD

- Ruff and Black (strict) → PyTest with coverage on Python 3.10 / 3.11 → Phase 2 pipeline → Docker image build → container smoke test (/health 200 with all models loaded, /predict, /metrics).
- Pushing the image to a registry on main is an open question and not part of Phase 2 yet.

Done when: the workflow is green on main and the smoke test passes.

11. Honest evaluation

Earlier headline numbers (0.97 accuracy / 0.95 F1) came from duplicate questions leaking between train and test. After de-duplication, and on real RVCE papers the models never trained on:

Model	In-domain test	RVCE held-out	Majority baseline
Difficulty — LR (Production)	acc 0.92 · F1 0.66	acc 0.41	0.46
Difficulty — XGBoost (Staging)	acc 0.95 · F1 0.69	acc 0.46	0.46
Bloom's level	acc 0.58	acc 0.24	0.47
CT skills (multi-label)	F1-micro 0.70	F1-micro 0.36	—

RVCE difficulty and CT labels are LLM drafts pending faculty review, so these numbers are reported rather than used as promotion gates. Faculty-labelled RVCE data is the main lever for improving them.